LEARNING MANAGEMENT SYSTEM

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1. INTRODUCTION:

1.1. Purpose:

This Document serves to specify all the User Requirement Specifications (URS) that need to be addressed by Design (Functional specifications), Configuration, Testing, Installation, Operational and Performance Qualification for the Learning Management System.

The requirements specified in this document will be verified in Installation / Operational and Performance Qualification tests by Alfa Biomed India Private Limited, Pune.

The key objective is to provide a process control system with the required functionality and flexibility while simultaneously meeting the requirements of ISPE GAMP 5, cGMP, FDA's 21 CFR part 11 and Alfa Biomed India Private Limited, Pune policies and procedures.

1.2. Scope:

The scope of this document is limited to 'Learning Management System' to be procured at Alfa Biomed India Private Limited, Pune.

This URS Document shall outline the baseline user requirements for the Learning Management System software.

The Validation of this system shall include the data control and application that cumulatively fulfill the requirements of this document and cGMP.

2. SYSTEM OVERVIEW:

The system must be able to interface with users having moderate expertise and experience in handling this system. The frequency may vary from User to User depending on the function they are handling.

This User Requirements Specification (URS) defines the requirements of the "Learning Management System", basically to generate the database to provide Data security, Data traceability and tracking functions to be carried out for management of trainings, the data on

which the system will operate, and the operating environment.

This includes the requirements associated with User's daily Business activity and supported processes (including output requirement, safety, access control, use of electronic signatures, validation messages/pop ups, data inputs, review and approval process).

URS ID / Section No.	User Requirement Description
2.1.	**GENERAL REQUIREMENTS**
2.1.1.	**Name of System / Application :** Learning Management System software module
2.1.2.	**Intended Use of System**
2.1.2.1.	The digitization of the training ensures the following prospective:
2.1.2.2.	Training process implementation and monitoring across all functions viz Quality Assurance, Quality Control, Production, Engineering, Human resources, warehouse, information technology, contractual employees, external training and etc..
2.1.2.3.	Automation of regulatory compliance to facilitate accurate auditing of various training related transactions online.
2.1.2.4.	To ensure timely alert and reminder for schedules as well as period trainings and tasks.
2.1.2.5.	To provide complete traceability and audit transactional data executed in system. Security of training data expected to have security implemented over who can view/modify and access trainings and maintain schedules for compliance under the stipulations of US FDA's 21CFR Part 11: Electronic Records: Electronic Signatures as well as EU Annexure 11 for security.
2.1.3.	**Description of system**
2.1.3.1.	The dedicated systems shall be made available for the installation and sufficient memory space will be made available for the configuration of application
2.1.3.2.	The generated data shall be store in server
2.1.3.3.	The system shall be fully integrated with DMS application
2.1.4.	**Product Class to be used**
2.1.4.1.	General category of Product shall be manufactured in the facility, hence generalized training activity shall be conducted

2.1.4.2.	There will be Total 7 Manufacturing lines at Site amongst which 5 are dedicated to OSD and 02 lines are dedicated to Oral Liquid manufacturing.
2.1.4.3.	In the facility the minimum 300 employees shall be working in different areas in which permanent, contractual as well as few vendors working for defined project shall be considered to access the system including Classroom, ILT, WBT, Blended sessions, External Webinars, Vendor arranged trainings, training given to vendors, External trainers given trainings (but not limited to)

3. SYSTEM REQUIREMENTS:

This section shall state the requirements in which the system must operate. Every requirement must have a unique identifier. Requirements will be as follows.

URS ID / Section No.	User Requirement Description
3.1	**Security / Regulatory Requirements:** This section shall specify all security specifications that are to be followed. This should include any special business control and data security requirements of the system. These should include protection from accidental or malicious access, use, modification, destruction, or disclosure of the system.
3.1.1.	System should not allow creation of duplicate username.
3.1.2.	In system, administrator shall have option to reset user's password when required.
3.1.3.	System should have password reset period and it shall be as per Alfa Biomed policy.
3.1.4.	System should have option to disable user account option in application.
3.1.5.	System should not have option to delete the user account.
3.1.6.	Verify that If application has option to delete the user account, then it should be recorded in audit trail.
3.1.7.	Application should have user groups and assigned privileges for each user group as per

URS ID / Section No.	User Requirement Description
	Alfa Biomed requirement
3.1.8.	System administrator privileges are given to Alfa Biomed IT and same is captured in Privilege matrix (wherever applicable)
3.1.9.	System date & time should be auto synchronized with Alfa Biomed Server for network connected systems *(Time server or Domain server)*
3.1.10.	In case of server base system, client system date & time is captured from server
3.1.11.	Application shall not store any of its data into windows public folders like profile, temp, event log etc.
3.1.12.	System supports for old version data retrievability and readability in newer version, in case of system upgrade. (Applicable during system upgrade/ reinstallation)
3.1.13.	Application allows for backup of electronic data including audit trail.
3.1.14.	Procedure is in place for periodic testing of data restore to ensure data accuracy
3.1.15.	Audit trail shall record all the activities of all users irrespective of type of the user (like operator, manager etc.,) including administrator
3.1.16.	Application shall store Audit trail data.
3.1.17.	System date and time shall be controlled and not editable by user
3.1.18.	Application has no option to disable the recording of electronic data
3.1.19.	Electronic data shall not be open for modification within the application
3.1.20.	Electronic data shall not be open for modification outside the application *(using external tools like notepad from data folder)*
3.1.21.	Electronic data shall not be deleted within the application
3.1.22.	Electronic data shall not be deleted outside the application (from data folder)

URS ID / Section No.	User Requirement Description
3.1.23.	Application shall store the raw data automatically into the default data folder without any operator's decision
3.1.24.	User should not have "*Reset*" or "*Abort*" or similar option, which stops the current activity without storing the data.
3.1.25.	Application should have facility to maintain data in year wise folder structure for ease of access control and archive (as applicable)
3.1.26.	In case of server-based application, there shall not be any impact of network failure during storing of data.
3.1.27.	Audit trail shall have 3Ws *(Who, When and What).*
3.1.28.	Audit trail data shall not be open for modification within the application
3.1.29.	Audit trail data shall not be open for modification outside the application *(using external tools like notepad from data folder)*
3.1.30.	Audit trail data generated in the system shall not be able to delete within the application
3.1.31.	Audit trail data generated in the system shall not be able to delete outside the application (from data folder)
3.1.32.	Application shall not have option to disable/pause the recording of audit trail
3.1.33.	Application shall accept only valid data within their acceptable limits (as applicable)
3.1.34.	Application shall not allow to proceed without mandatory data fields entered
3.1.35.	System should perform all the actions in the correct sequence where sequenced steps are required.
3.1.36.	Application data is not altered and/or meaning not changed during the exchange process where system is interfaced with other systems **(Applicable for interfaced systems)**

URS ID / Section No.	User Requirement Description
3.1.37.	System should not allow to Login with invalid username and valid password.
3.1.38.	System should not allow to Login with valid username and invalid password.
3.1.39.	System should not allow to Login with invalid username and invalid password.
3.1.40.	System should allow to Login with valid username and valid password.
3.1.41.	System should not allow to set the password having length less than 8 characters.
3.1.42.	System should allow to Set the password length equal to minimum password length ‘ ’ characters.
3.1.43.	System should allow to Set the password length more than minimum password length ‘8’characters.
3.1.44.	Users shall have the option to change the password when required
3.1.45.	System shall have provision for individual user account
3.1.46.	System should not allow to Login with valid username and invalid password for 3 more times (Configurable). The system should have Unauthorized login attempts.
3.1.47.	The system should keep the application idle for 15 minutes (Configurable).
3.1.48.	Procedure is established that holds individuals accountable and responsible for actions initiated under their electronic signatures.
3.1.49.	E-signature certification has been submitted to the FDA, stating their intent to use electronic signatures.
3.1.50.	System should require both components [username and Password] of the electronic signature when signing a record for the first time.
3.1.51.	System should require one component i.e. password (at a minimum) which is designed and can be executed by genuine user only for subsequent electronic signatures during a continuous session

URS ID / Section No.	User Requirement Description
3.1.52.	E-Signature shall appear in reports like batch report, certificate of analysis *(where applicable)*
3.1.53.	Electronic signature shall appear on a printed report/record, information (minimum requirements): - Printed name of signer (Who) - Date & time stamp of signature (When) - Associated meaning of the signature such as Review, Approval. (What was done)
3.1.54.	Electronic signature shall appear in audit trail (minimum requirements): - Printed name of signer (Who) - Date & time stamp of signature (When) - Associated meaning of the signature such as Review, Approval. (What was done)
3.1.55.	Validation of the system should be planned and conducted as per approved validation protocol using pre-approved test procedures.
3.1.56.	Procedure should be established for System operation User & System administration CSS change control Incident and CAPA management Periodic Review.
3.1.57.	Procedure for formal risk assessment and reporting should be in place throughout the life cycle of the system.
3.1.58.	System shall generate accurate and complete copies of record in both human readable and electronic form.
3.1.59.	Procedure should be established to ensure appropriate qualifications, experience and/or training to personnel.
3.1.60.	Procedure should be established for supplier assessment / vendor audit/formal

URS ID / Section No.	User Requirement Description
	agreement.
3.1.61.	System should allow qualified person for recording certification and batch release, in case system is used for batch release & it is performed using electronic signature.
3.1.62.	There should be availability of business continuity plan/activities.
3.1.63.	Password should be complex (at least alphanumeric characters).
3.1.64.	Generic user account should not be available in system

4. TECHNICAL REQUIREMENTS

URS ID / Section No.	User Requirement Description
4.1.	**TECHNICAL REQUIREMENTS**
4.1.1.	**Process Control Requirements**
4.1.1.1.	The system-based design control and data handling system shall be in place over multiple training activities are arranged during all operations involved in R&D scale to commercial scalability which includes Induction, On Job, Remedial and refresher trainings.
4.1.1.2.	These User Requirements will be used to generate functional requirements for Learning Management System (LMS) and will provide a basis for the creation of acceptance criteria for the validation of system
4.1.1.3.	Learning Management System software shall be useful to generate, execute and store the data electronically, ensure data integrity and prevent data manipulation.
4.1.1.4.	Provision for authorized users to review historical data.
4.1.1.5.	The system shall have an authorization mechanism to ensure that access to data and functions in the application can be used based on job responsibility.

URS ID / Section No.	User Requirement Description
4.1.1.6.	The system should facilitate all the options to the users based on the Roles/privileges which are assigned to them.
4.1.1.7.	The LMS shall display the contents which are assigned to User based on JD.
4.1.1.8.	The system should hide the Training material which has expired and should be made unavailable for any of the user except administrator / QA head (in obsolete folder).
4.1.1.9.	The system should only allow the administrator to make the active documents to make it inactive. The system should take care entire training life cycle of each document right from Under training to completion of training on content.
4.1.1.10.	The system shall have user interface available in Multilingual language. Marathi & Hindi language shall be considered as these are the local languages.
4.1.1.11.	The System for Error Handling/ Alarm Requirements shall be documented as applicable i.e., Alarms/ Errors message/ warning message.
4.1.2.	**Application of the system**
4.1.2.1.	Learning Management System (LMS) is useful in regulating the Document Training content flow from draft to under training to defined level of training completion in system based in phased manner.
4.1.2.2.	Learning Management System (LMS) shall provide controls on Training content prepared and executed end to end in Digital platform
4.1.2.3.	Learning Management System (LMS) Shall be integrated DMS simultaneously to collect and store documents. The DMS and LMS shall work closely to monitor entire life cycle development of document whereas rest of applications also have business process in day-to-day operations
4.1.2.4.	The entire life cycle development of all training activities generated and executed in the facility shall be triggered through LMS system
4.1.2.5.	Training Management System (LMS) can be proposed for the following Document Types:

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	✓ Policies like Quality, EHS, IT ✓ Quality Manual & Training Manual, ✓ SOP's and Related Annexure as per TNI ✓ Protocol based studies ✓ Basic GMP & Personal Hygiene to Induction to Periodic refresher trainings ✓ Remedial and External Trainings ✓ Classroom trainings, ILT, WBT and Blended sessions etc.
4.1.3.	**ISO Grade requirement of the surroundings:** NA
4.1.4.	Utilization of System on Hours basis: The System should be implemented by considering that it should be up and run 24/7.
4.1.5.	**Spares and Accessories:**
4.1.5.1.	Supplier shall be responsible for replacement, at no charge, of all faulty electronic components or material of defective workmanship for 12 months after installation List of recommended spare parts with code no & ordering format shall be given by supplier considering 2 years of smooth working of machine.
4.1.5.2.	System connectivity should be ensured through networks even the connection links are in longer distance. System should be available for the users who are in different locations of the site; preferable on mobile phones.
4.2.	**OPERATIONAL REQUIREMENTS**
4.2.1.	**Data & Operational Requirements**
4.2.1.1.	**Set Security Profile**: The Administrator user shall set the complete security profile for the system commensurate with the company's policy involving electronic documents and signatures. Modifications to the set security profile shall be done if required and shall be addressed in Audit trail.
4.2.1.2.	**Register New Role**: New roles shall be registered, by assigning each role a name,

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	based on the category (nature of work), role name (indicative of the designation or position), and Level (position in the hierarchy of line of control). New role registration shall enable to register the various roles of the users in the facility
4.2.1.3.	**Set** Profile: There shall be Profile to set the rights and privileges authorized for a particular registered Role. All users of a given Role shall have the same functions and privileges as set for that Role (unless restricted in the User Profile Setting of individual users). The Profile Setting is limited to a particular Role Name assigned to a particular Plant or Master.
4.2.1.4.	**Register New User**: To register and manage users, along with their particulars as Master Data Entities.
4.2.1.5.	**New User Sign Up & Activation**: New User Sign Up shall enable the newly registered user to sign up, provide personal particulars, and submit for activation of his / her account.
4.2.1.6.	**Plant Assignment**: Plant Assignment (*OSD or Oral Liquid for few Users*) is to assign a registered User to a Plant i.e., users may be assigned to new plants or may be removed from previously assigned plants.
4.2.1.7.	**Set User Profile**: User Profile is a set of rights and privileges authorized for a particular registered User. The User Profile Setting shall be Subset of Global Profile Setting allowed for Role to which the user belongs.
4.2.1.8.	**Login Password**: There shall be facility to reset a user's login password, although, only upon the request of a user, in case the user had forgotten his / her login password and wanted a reset of the same.
4.2.1.9.	**Personal Settings**: There shall be provided a provision for each Individual user to view/selectively modify his/her personal particulars that are provided at the time of the user sign up.
4.2.1.10.	**Standard Reasons**: There shall be reasons needed to be stated for actions such as

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	approval, return, drop, modification, status change etc. To maintain consistency and to reduce tedious and error prone repetitive typing, known common reasons may be standardized. The Standard Reasons Registration facility is to enable creation of a ready bank of Standard Reasons for retrieval when needed. The status of already registered Standard Reasons can also be changed to “Inactive” as may be warranted due to typing errors or obsolescence.
4.2.1.11.	**User Session Log**: The login and logout details of the users shall be tracked through the user session log
4.2.1.12.	**View User Messages**: There shall be an option to View user messages for the authorized user, to see the request details for password reset, reactivation etc.
4.2.1.13.	**User Account Status:** There shall be a provision for the viewing the Active and Deactivated user details through user account status.
4.2.1.14.	**Training material library:** There shall be provision for creating the category/ type of the training material for easy identification of the material type and provision shall be provided to register the document details under the specified type of the document
4.2.1.15.	**Document Reading:** Irrespective of scheduled or unscheduled or of Interim session type, this option shall be used when no evaluation is required, and trainings can be considered carried out when one or more documents of a given course are to be only read by the identified associates
4.2.1.16.	**Course Session**: Provision shall be provided for creation of three types of course sessions namely scheduled, unscheduled and interim sessions
4.2.1.17.	In all the session types, option to carry out the session in an On-Line or an Off-Line mode shall be available. For On-line mode of trainings, option shall be provided for the trainees to answer the question papers online.
4.2.1.18.	Option shall be available to select the missed questions analysis for every online session with examination. During the selection of the user for the session, option to

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	view the trainee's last session status details such as Subgroup to which the user belongs to, when the concerned course was last Attended On, the status of the same, Last Session Course Version, Current Session Status (If any) and Current Session Course Version shall be available.
4.2.1.19.	No option shall be available to select those trainees whose job responsibilities are not defined in the given plant. Option to select the method of training as one of 'Internal Classroom', 'External', 'Shop Floor', 'Document Reading' or 'Completed Training' shall be available
4.2.1.20.	**Reports:** Reports for the uploaded training material shall be created to ensure the user to identify the documents that need to be reviewed. There shall be mail escalations once the review extension date has been crossed.
4.2.1.21.	**Question Paper Preparation**: Option to prepare a question paper for each batch of a course session shall be available in the respective plant, for the authorized user. The option to prepare a question paper shall be available for only those training sessions for which examination is a marked as required and for which topics questionnaires are available. Option to select the Evaluation method as either 'System Evaluation' or 'Manual 'shall be available. In case of System evaluation mode, the system shall automatically evaluate all the answers and also declare the trainee Qualified or to be retrained based on the result. No intervention of an evaluator is required. In case of Manual mode option will be available to select an Evaluator at the time of record attendance, who shall then carry out the evaluation after trainee responds to the Question paper.
4.2.1.22.	Multiple search options shall be provided with Training document type, name, review period, etc.
4.2.1.23.	Electronic records of Training activity shall protect unauthorized access to data & its manipulation.

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4.2.1.24.	The system shall be able to accelerate the operational speed and security.
4.2.1.25.	The system shall be able to establish control either parallel, serially up to the required or specified hierarchy level as it can configured by user for particular transaction.
4.2.1.26.	All the transactions shall be done as per the modified details in the system.
4.2.1.27.	The system shall be able to control the filling all the mandatory fields then only the forms shall be submitted.
4.2.1.28.	The system shall have automatic numbering system for the document created which shall be incremental with respect to the project / procedure. The department name, type of document and sequential numbering shall be taken into consideration while assigning numbers
4.2.1.29.	The system shall be able to shoot the confirmation message for all the submissions.
4.2.1.30.	The system shall be able to control that all the modified details should be approved by the authorized persons.
4.2.1.31.	The system shall display the appropriate preloaded template for the type of Training content / type as chosen by the creator.
4.2.1.32.	The system shall ask the Reasons/ Comments/ Remarks for all the modifications and that shall be captured.
4.2.1.33.	The system should send an E-mail alert to the concerned after assigning of TNI or allocation of new training content.
4.2.1.34.	Each entry to submit the workflow i.e. assessment, course reading shall be requested for electronic signature and separate record shall be generated on request from the system.
4.3.	**DOCUMENT CONTROL REQUIREMENTS**
4.3.1.	The system should inform the originator if the Training on specified Document / SOP has been completed
4.3.2.	The system should have the facility to provide the completion date of Training with

URS ID / Section No.	User Requirement Description
	percentage as well as details of pendency
4.3.3.	Plant **Wise User List** The list of users assigned to a given plant shall be viewed by the plant wise (OSD and OLD) user list provision
4.3.4.	Configurations The initiation/number of approval levels for each of the transaction shall be configured as per the requirement and the criticality of the transaction. The E-Sign shall be the approval confirmation
4.3.5.	**Import Users** Importing the users at one time as a bulk user registration and enrolment shall be available to reduce the timeline lag for the activity.
4.3.6.	**Mail Settings** The mail settings option shall be provided for enabling the events to which mail notifications need to be sent to the concerned persons.
4.3.7.	**Dash Boards** Details the members of the team and their independent training status in addition to this my training program status and plant training status for the current month also shall be provided
4.3.8.	**User Groups & Subgroups** There shall be provision for assigning the user to register Subgroups, Groups and assign the users to respective Subgroups and also assign Job Responsibility. This signifies training identification for the given role and groups of users. The modifications/ changes to the any sub group/ group shall be notified in the audit trail.
4.3.9.	**Job Responsibility** Job responsibilities shall be defined for every user at every level and the responsibilities defined for the concerned user groups will by default be populated as the respective user's responsibilities. User acceptance shall be available for the job responsibilities and provision shall be provided to select the multiple alternate responsible person.
4.3.10.	**Reports**: To give the detailed information about the groups, subgroups and Job Responsibility of employees there shall be reports for critical transactions; all the reports shall be available for the print.

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4.3.11.	Group wise user report: Detailed
4.3.12.	Job Responsibility Report: To view the latest and the active revision job Responsibility particulars of an employee assigned to that plant
4.3.13.	Job Responsibility History Report: To view all the detailed change information/revision history of the Job responsibilities of every employee
4.3.14.	**Internal Classroom training / ILT**: Irrespective of scheduled or unscheduled or of Interim session type, all training sessions that are conducted in a class room environment where a trainer explains the subject matter and the trainer also belongs to the organization can be registered under this type of Training.
4.3.15.	**External Trainings**: Irrespective of scheduled or unscheduled or of Interim session type, typically this mode of training shall be used when an external trainer provides the training and evaluations of the trainees.
4.3.16.	**Blended Sessions**: The scheduled training session type, typically this mode of training shall be used when any new Training content gets implemented and ILT as well as WBT both sessions are accepted based on percentage of employee available for ILT and rest employees goes for WBT training.
4.3.17.	**On-the Job**: Irrespective of scheduled or unscheduled or of Interim session type, this type of trainings shall be conducted by means of practical experience on the selected subject over a period of time for the selected employees.
4.3.18.	**Document Reading:** Irrespective of scheduled or unscheduled or of Interim session type, this option shall be used when no evaluation is required and trainings can be considered carried out when one or more documents of a given course are to be only read by the identified associates
4.3.19.	**Completed Trainings:** Irrespective of scheduled or unscheduled or of Interim session type, option shall be provided to register any training already carried out or those trainings carried out but session particulars could not be transacted in the system for

URS ID / Section No.	User Requirement Description
	any reason. Though the session is already conducted, the particulars of the same such as the session creation, attendance etc. shall be carried out in the system postdated but the evaluation and other particulars shall be available in the system.
4.3.20.	**Attendance Record:** Option to record the attendance for Batch will be available for all methods of training except for the Document Reading method. Option to record the attendance after taking the Attendance Record print of the respective Batch will be available. During the attendance recording, option to provide the actual start time and end time of the session will be available. Option shall also be available to upload the scanned copy of the attendance record.
4.3.21.	**Induction Training:** At the time of employee registration, option shall be provided to decide if the given employee requires induction training. All such users who are marked for induction training shall have to undergo induction trainings before the general courses are available for them for training. Option shall be available for defining the induction course trainers into a specific group. Only those users who are in the group can be selected as a trainer for the induction courses
4.3.22.	**Reports:** The following reports shall be provided for the effectiveness check of the training system (But not limited to): Qualified Trainers List Training Card Training Evaluation Record Employee Training History Individual Employee Report Employee Status Report for specific course/topic Training Schedule Report Attendance Report Document Reading Pending

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	Question Bank Report Terminated Course sessions Course Sessions Report Training proposal Scheduled Courses Pending list Topic-Course Report Course sessions (complete) report Course List Report Induction training assessment
4.4.	**FUNCTIONAL REQUIREMENT:**
4.4.1.	Option to prepare Portrait/Landscape header/ Footers in single Training material shall be provided.
4.4.2.	Option to prepare Training Materials like questionnaire, master answer key sheet for training materials.
4.4.3.	Department name should be dynamic based on user login.
4.4.4.	Training Materials Revision/ change History shall be provided
4.4.5.	System should have option to configure a process for retention of regulated Records
4.4.6.	System shall be available with delegate option considering the unplanned leave of employee.
4.4.7.	System should have option to create periodic training schedules for all departments based on content wise requirement
4.4.8.	System should have option to perform periodic review of training materials
4.4.9.	System should have option to view/review audit trail report for every action user performs for all objects under each module
4.4.10.	**User Session Log**: The login and logout details of the users shall be tracked through the user session log.

URS ID / Section No.	User Requirement Description
4.4.11.	**View User Messages**: There shall be an option to View user messages for the authorized user, to see the request details for password reset, reactivation etc.
4.4.12.	**User Account Status:** There shall be a provision for the viewing the Active and Deactivated user details through user account
4.5.	**GMP REQUIREMENTS (GAMP/21 CFR Part 11, USER ACCESS MANAGEMENT)**
4.5.1.	The system shall have an authorization mechanism to ensure that access to data and functions in the application can be used based on job responsibility
4.5.2.	The system shall provide user access based on Employee job responsibility.
4.5.3.	The system / administrator must have the facility to provide different level of access for different projects for every member
4.5.4.	The system should not allow any folders or documents to be deleted from the system, and only administrator / QA Manager should have such access
4.5.5.	The application shall not display the passwords.
4.5.6.	The system Shall be able to limit access to authorized individuals through a unique personal combination of user ID and password
4.5.7.	The system / administrator must have the facility to provide different level of access for different projects for every member.
4.5.8.	Administrators shall not use passwords derived from commonly used words or phrases
4.5.9.	The system shall inform only user / administrator about password create or change by e-mail
4.5.10.	Users changing a password via a system command or screen must prove knowledge of the current password
4.5.11.	The application shall not allow the reuse or reassignment of e-signatures
4.5.12.	The system shall be able to link (via technology, not procedures) e-signatures to their corresponding electronic records

URS ID / Section No.	User Requirement Description
4.5.13.	The application shall be able to produce the printed name, date, time and signature meaning in every human readable form of the electronic record (e.g. all screens and printed reports)
4.5.14.	Records produced by the system or records used as system input shall be kept in readily retrievable electronic format as required in the company Record Management Schedules. This applies to both current and archived documents.
4.5.15.	The application shall derive the date and time stamps in a consistent way in order to be able to reconstruct the sequence of events.
4.5.16.	System should provide option or allow for backup of data in secure location i.e. cloud based platform.
4.5.17.	System must keep a log of unauthorized attempts to access system resources or sensitive information.
4.5.18.	Audit trail must be printable. Entries must be recorded automatically when a transaction occurs in audit trail.
4.5.19.	Audit Trail report shall be generated for every transaction done in the LMS
4.5.20.	Audit trails shall not be editable & reasons are also logged for changes.
4.5.21.	The system shall provide role based on authorized login
4.5.22.	Computer Generated audit trails shall be maintained to record the date and time that electronic records are modified or created
4.5.23.	Re-use of already assigned user ID should not be accepted, even if disabled
4.5.24.	Every user shall be unique & cannot be deleted except the Administrator
4.5.25.	Password ageing: Provision of password change at regular intervals
4.5.26.	**Audit Management system Details:** Any changes regarding quality impact should have historical record of data and it should have a provision that user can configure an audit trail for particular flow as per user requirement and Audit Trail should be available for all modifications including administrative changes and user login/logout

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	details. Audit Trail cannot be disabled by any Role or User.
4.5.27.	**System Failure Action/Messages**
4.5.27.1.	If user tries to access the system software with wrong password, then it should not allow more than three attempts. If user tries to enter wrong data types then system should not accept that values and should generate error messages.
4.5.27.2.	If user tries to enter wrong data types, then system should not accept that values and should generate error messages.
4.5.28.	**Data Integrity requirements:**
4.5.28.1.	To minimize the risks, the LMS shall ensure that there are appropriate built-in checks for the correct and secure entry and processing of data when data is exchanged electronically with other systems.
4.5.28.2.	The application shall be able to include validation checks that GxP data are not altered in value and/or meaning if they are transferred to another data format or system.
4.5.28.3.	The application shall be able to maintain continuous adherence with ALCOA Plus Principles.
4.5.28.4.	'ALCOA' defines that data should be Attributable, Legible, Contemporaneous, Original, and Accurate. In addition, 'ALCOA+' guidance recommends that data is also Complete, Consistent, Enduring, and Available.
4.5.29.	**Electronic record requirements**
4.5.29.1.	The application shall be able to generate accurate and complete copies of GxP electronic records in human-readable form
4.5.29.2.	The application shall manage diverse master data such as material basic data in master data tables as electronic records
4.5.29.3.	The application shall be able to manage the documents as an electronic record. It shall also be able to manage the equipment status as an electronic record
4.5.30.	**Electronic Signatures Requirements:**

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4.5.30.1.	Electronic Signature means a computer data compilation of any symbol or series of symbols executed adopted or authorized by an individual to be legally binding equivalent of the individual's handwritten signature.
4.5.30.2.	System should have an option for the security Requirements of Login Complexity, User Access Rights, Electronic Signatures, and Logical Security
4.5.30.3.	The application shall be able to contain information associated with the signature that clearly indicates the identity of the signatory, the date/time when the signature was executed and the reason of the signature.
4.5.30.4.	The application shall be able to link their respective electronic records to ensure that the signatures cannot be cut out, copied, or otherwise transferred to falsify an electronic record by ordinary means.
4.5.30.5.	The system shall be able to identify each electronic signature as unique to one individual, not allowing it to be reused by or reassigned to anyone else.
4.5.30.6.	The system shall prevent the deletion of GxP records.
4.5.30.7.	System should log electronic signatures along with name of signer, date/time stamp and reason/meaning of the signature.
4.5.30.8.	System should enforce unique electronic signatures for critical user actions such as create, modify, initiate, review, approve, accept, reject, and return.
4.5.30.9.	System should securely link the electronic signature with respective records
4.5.30.10.	System should allow for configuring mandatory prompt for user id / password, during modifying/confirming identified records.
4.5.31.	**Audit Trail Requirements:**
4.5.31.1.	The application shall be able to provide secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create modify or delete electronic GxP records.
4.5.31.2.	The application shall be able to maintain under audit trail control.

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4.5.31.3.	The application shall be able to provide a reason for modifying the GxP records where appropriate.
4.5.31.4.	The application shall be able to ensure that audit trail information cannot be deleted or modified. The application shall be able to ensure that changes to the GxP records shall not obscure previously recorded information.
4.5.31.5.	The application shall be able to ensure that audit trail information shall be retained for a period at least as long as that required for the related electronic GxP records and shall be available for review and copying Time and date stamps for Electronic Batch Record that span different time zones shall be implemented with a clear understanding of the time zone reference used. Shall allow time zone tracking; however, the Electronic Batch Record is supposed to be Installed locally.
4.6.	**POWER FAILURE AND RECOVERY MANAGEMENT**
4.6.1.1.	System should restrict willful or accidental loss, damage, or unauthorized change.
4.6.1.2.	In case of failure, system should allow the user to restore it to the latest working condition to reduce the data loss in case of failures.
4.6.1.3.	System should work in minimal abnormal conditions related to power including voltage, amperage, filtering, loading, earthling protection, uninterruptible power supply (UPS) etc.
4.6.1.4.	In case of failure, system should allow the user to restore it to the latest working condition to reduce the data loss in case of failures.
4.6.1.5.	System should be In extreme cases, system should tolerate minimal abnormal physical conditions due to temperature, humidity, external interference etc.
4.6.1.6.	The application shall be able to file the data in a way that enables the archiving of the data.
4.6.1.7.	The application shall ensure data consistency even in case of a program crash (use

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	of transactions).
4.6.1.8.	The LMS shall be able to administrate the information under the concept of redundancy and high availability.
4.6.1.9.	The recovery time objective (RTO) : Targeted amount of time after which the LMS shall back in operation after a disaster should be minimal i.e 15 Min. Disaster recovery shall be available.
4.6.1.10.	There shall be procedure available for the following activities Periodic backup of system data Periodic Restore of system data Business continuity Disaster recovery System administration
4.7.	**DOCUMENTATION REQUIREMENTS:**
4.7.1.	**Installation, Commissioning, and documentation:**
4.7.1.1.	Installation & commissioning followed by training to operating personnel & staff with no extra cost shall be the responsibility of the supplier.
4.7.1.2.	The supplier shall provide the required documentation (compliance to cGMP) at the desired stage of operation/manufacturing for preliminary review. After approval the final documents shall be provided. The Supplier shall provide documentation reflecting "as-built" condition with final delivery. . All validation deliverables . Quality Instance and production Instance. . Operation manual. . Complete drawings with details required for Utility connections.
4.8.	**SAFETY**
4.8.1.	**Security and Integrity Requirements**

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4.8.1.1.	Server should have Windows server 2008 R2 above based operating system with proper administrator rights defined. So, it should not allow the following. Date and time modification. Server shutdown Server start-up
4.8.1.2.	System software should have its own security module for login/logout, accessing different features of the software.
4.8.1.3.	Software should have multilevel security configuration for the application client, server, and database administrator, so it should not allow access in any other means.
4.8.1.4.	Restriction for the unauthorized person to client module and server database.
4.8.1.5.	Security module should be provided to create a database for individual users access rights of the other system components.
4.8.1.6.	Operating system security should be configured to protect the system from unauthorized access & thus to prevent modifications in data files, date/time, start-up/shutdown, etc
4.8.1.7.	It should meet all clauses as per 21 CFR Part 11.
4.8.1.8.	**Reset Login Password:** There shall be facility to reset a user's login password, although, only upon the request of a user, in case the user had forgotten his / her login password and wanted a reset of the same.
4.8.1.9.	**Personal Settings:** There shall be provided a provision for each Individual user to view/selectively modify his/her personal particulars that are provided at the time of the user sign up
4.8.1.10.	**Standard Reasons:** There shall be reasons needed to be stated for actions such as approval, return, drop, modification, status change etc. To maintain consistency and to reduce tedious and error prone repetitive typing, known common reasons may be standardized. The Standard Reasons Registration facility is to enable creation of a

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	ready bank of Standard Reasons for retrieval when needed. The status of already registered Standard Reasons can also be changed to “Inactive” as may be warranted due to typing errors or obsolescence.
4.8.1.11.	**Audit Trails:** The audit trails will keep chronological accounts of all the important events and actions with time stamps of the events and actions with electronic signatures of the users who caused, performed and / or authorized those events and actions, along with reasons / remarks recorded there for. The audit trails will not only help track and trace the events but also fulfil stringent regulatory requirements. The audit trails give the system related events with signatures (login usernames) of the users concerned and time stamps along with remarks made by the concerned user. The audit trail shall be viewed by the authorized personnel so that the system compliance is monitored from time to time.
4.8.1.12.	System should allow administrator to define/modify system security controls as per the bellow password policy with e-signature process.
4.8.1.13.	System Should allow the Admin to create/modify/deactivate the role with required privileges.
4.8.1.14.	System should provide access only to authorized users through user identity and password combination.
4.8.1.15.	Password complexity: Password length should be minimum 8 characters and alpha numeric. Password should have session expiry period setting and configurable. System should mask password entry
4.8.1.16.	System should enforce unique user IDs. System should allow authorized user to set user account status and set user account status as active / inactive
4.8.1.17.	Proper session handling mechanism should be available in the system
4.8.1.18.	The system must allow system administrators to specify a request for password change during initial login. System should allow the users to change their password

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	when password expires. System shall prompt the user with change password when user first login.
4.8.1.19.	The system must lock user accounts after 3 times of incorrect login attempts
4.8.1.20.	System should allow the user to reset password when user forget password.
4.8.1.21.	The system shall have change password and user deactivation facility.
4.8.1.22.	Computer Generated audit trails shall be maintained to record the date and time that electronic records are modified or created.
4.8.1.23.	No users shall be permitted re-use of their last three passwords
4.8.1.24.	Re-use of already assigned user ID should not be accepted, even if disabled
4.8.1.25.	Every user shall be unique & cannot be deleted
4.8.1.26.	**Password ageing:** Provision of password change at regular interval.
4.9.	**OTHER REQUIREMENTS**
4.9.1.	**Timeline**
4.9.1.1.	A minimum of two weeks shall be allowed for the user to review and to comment and/or approve test specification.
4.9.1.2.	The user shall notify the supplier of the length of runs required, special cases required, and any other unique test requirements two weeks in advance of the start of testing.
4.9.1.3.	System should be open to allow further enhancements if any as decided by Management
4.9.1.4.	System should be implemented by considering that it should be durable for long term with high efficiency in the performance
4.9.2.	**Validation:**
4.9.2.1.	Vendors need to provide compliance certificate. Vendor needs to be supported for Functional Specification Preparation & approval Functional Risk Assessment Preparation & approval,

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	Software design, configuration, testing and release UAT 21 CFR Part 11 Preparation and approval. Installation and operational qualification Quality Instance and production Instance.
4.9.2.2.	**Statutory:** Services supplied or installed against this specification shall meet all local legislative requirements. - cGMP –Title 21 CFR, Part 210 and 211. - GAMP
4.9.2.3.	**Regulatory:** It should meet all the regulatory expectations which are mentioned in section cGMP section.
4.9.3.	**Delivery Schedule:** Vender to submit bar chart for various activities at the time of offer**.**
4.9.4.	**Installation Site:** All final documents shall be shipped with transmittals that identify them as contractually required documents. All final documents and drawings shall reflect "as-built" condition**.** All documents shall be supplied with hard copies and electronic versions supplied in the format identified for each document.
4.9.5.	**Support** • Any service/ support requirements required by third party/ in-house shall be documented. • The vendor shall have system for continuous support throughout the life cycle of the system

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	• Maintenance contract shall be in place for the system. • Operation as well as maintenance training shall be conducted and managed by supplier with necessary documentation. • It also defines any non-functional requirements, constraints such as time and costs, and what deliverables are to be supplied. • **Start-up support:** Training • **Post Start-up support:** • **Technical support:** Telephone, Email • **User site support :** System improvements • **Patch validation**
4.9.6.	**Warranty**
	As per technical agreement **Testing And Warranty:** Software designed and configured must be tested and released Software designed and configured shall be guaranteed for trouble-free operation for the period and shall be guaranteed against faulty design, defective or failure from normal use during the period. In case of non-fulfilment of guarantee, vendor shall repair / replace such defective software supplied and installed. The vendor shall be liable for all expenses that occurred for such repair / replacement.
4.9.7.	**AMC** Trained service engineers must be deputed promptly \ Suitable commercial arrangements shall be made between both the parties for breakdown calls for after 12-month period.
4.9.8.	**Retirement Phase:** System should have proper retirement plan/ Withdrawal plan/Disposal and process

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	for the systems which are going to be retired/Withdraw based on the business requirement. System should have proper Withdrawal plan and process to de-activate users, disable interfaces, data reporting, Result analysis and support. System should have proper Disposal plan and process to maintain documentary evidence for retained actions, GXP records. System should have proper Data Migration plan when an existing system is replaced by new System.

5. Abbreviations:

LMS	Learning Management System
DMS	Document Management System
URS	User Requirement Specification
CFR	Code of federal regulation
ISPE	International standards of pharmaceutical Engineering
SAT	Site Acceptance Test
GAMP	Good, automated manufacturing Practices
USFDA	United States of Food and Drug Administration
WBT	Web Based Trainings
R&D	Research and Development
ID	Identity
ISO	International Organization for Standardization
QA	Quality Assurance
ILT	Instructor Lead Training

6. REFERENCE

Nil

7. REVISION HISTORY

Revision No:	Details of Review / Revision
00	New user requirement specification

8. APPROVAL

PREPARED BY

Name	Department	Sign and Date

REVIEWED BY

Name	Department	Sign and Date

APPROVED BY

Name	Department	Sign and Date